

ALA'ALDIN MEFLEH

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Mechanical Design Engineer specializing in Advanced Finite Element Analysis (FEA) and Structural Integrity. Expert in nonlinear simulations, contact modeling, and code-based design for high-pressure systems and complex steel structures. Proven track record in optimizing mechanical systems through rigorous numerical analysis and streamlining engineering workflows.

TECHNICAL SKILLS

- **Analysis:** Finite Element Analysis (FEA), Linear/Nonlinear Static Analysis, Piping Stress Analysis, Vibration & Thermal Loads, Seismic/Wind Analysis.
- **Software:** Ansys Mechanical, Caesar II, PV-Elite, Robot Structural Analysis, SolidWorks (Simulation & Modeling), KeyCreator, AutoCAD Plant-3D, CADWorx.
- **Codes:** ASME B31.1, B31.3, ASME BPVC Section VIII Div 1, API 650.

PROFESSIONAL EXPERIENCE

FLUCON | Steel Structure & FEA Engineer

Amman, Jordan & Riyadh, Saudi Arabia | Apr 2024 – Present

- **Advanced Simulation:** Lead the FEA of complex support systems and industrial platforms using Ansys Mechanical, focusing on nonlinear contact algorithms and bolt pretension.
- **System Integrity:** Conducted comprehensive stress analysis on high-pressure routing using Caesar II, utilizing thermal and dynamic load cases to optimize system longevity per ASME standards.
- **Pressure Vessel Design:** Engineered thermal buffer tanks in compliance with ASME BPVC Section VIII Div. 1 using PV-Elite.
- **Pressure Vessel Optimization:** Currently developing a "Master Model" in **SolidWorks** linked to input data sheets to automate the generation of 3D models and shop drawings for pressure vessels, ensuring rapid, error-free manufacturing.
- **Standardization:** Developing a client-facing Structural Catalog that integrates piping analysis load data into standardized support typologies.
- **Project Leadership:** Oversaw fabrication and production in Saudi Arabia (Oct–Dec 2025), resolving constructability issues and ensuring compliance with IFC drawings.
- **Process Automation:** Revolutionized the piping support design workflow by integrating CADWorx and CAESAR II. Developed a custom Excel-based script that automatically sorts maximum node loads and exports them directly into CAD layouts as tags, significantly reducing manual data entry and design lead times.

MISCo. Holdings | Mechanical Engineer

Amman & Saudi Arabia | Nov 2022 – Mar 2024

- **Engineering Analysis:** Examined client specifications and design inputs to propose optimal mechanical solutions using rigorous computations and software simulation.
- **Mechanical & Structural Design:** Designed and modeled **API 650 tanks**, pump baseplates, walkways, stairways, and various steel structures. Supervised the assembly and production process to ensure design fidelity.

- **Large-Scale Infrastructure:** Collaborated on a 30,000-ton district cooling plant, preparing P&IDs, GA drawings, material submittals, and compliance papers.
- **Piping & Electrical Supports:** Performed loading and stress analysis for a 755 m² test loop system with 70 custom pipe supports. Designed and coordinated 430 electrical cable tray supports for a 28-chiller cooling plant.
- **Heavy Lifting & Logistics:** Engineered a specialized spreader beam and drafted the lifting plan to safely offload a 61-ton chiller.
- **Procurement Support:** Managed material sourcing by preparing purchase orders, recommending vendors, and estimating delivery timelines in coordination with procurement.

JoVision | Mechanical Engineering Intern

Mafraq, Jordan | May 2022 – Oct 2022

- **R&D:** Analyzed the mechanical feasibility of tire inspection machinery and designed a 3D scanner mechanism.
- **AI Integration:** Utilized neural networks and AI to develop high-accuracy text recognition systems.

EDUCATION

Bachelor of Science in Mechanical Engineering | Jordan University of Science and Technology

GPA: 3.59/4.00 (Excellent) | Jul 2017 – Jun 2022

CERTIFICATIONS

- Pipe Stress Analysis (CAESAR II & AutoPIPE)
- Structural Steel Design Principles
- Pressure Vessel Design Masterclass (ASME VIII)

LANGUAGES

- **Arabic:** Native Speaker
- **English:** Highly Proficient